



AI Nutritionist

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Smart Nutrition Assistant Companion & Curator (S.N.A.C.C.)

- Machine Learning Model

- Ingredients + Dietary Preference → List of Ranked Recipes
- User experience - Fast, easy, intuitive
- Utility - Efficient, effective

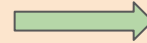
User Inputs:

- Pantry Inventory
- Dietary Preferences
(Vegan, Vegetarian,
Gluten Free, Dairy Free)



Recommendation Model:

- Content-based filtering
- Scoring Algorithm



Output:

- Top Recipe Recommendations



Problem Addressed - Impact of Food Waste on Environment and Economy

- 40% of food is wasted having a direct impact on the environment and consumption of natural resources (rescuingleftovercuisine.org)
- 19% of food waste is from retail, food service, and households (moveforhunger.org)
- Approximately \$243,864,931,645 in food waste per year from America, \$728.14 per capita (epa.gov)



Data Collection and Cleanup

INPUT (Dataset Used):

Extended Recipes Dataset: 64K Dishes
<https://www.kaggle.com/datasets/wafaaelhusseini/extended-recipes-dataset-64k-dishes>

This dataset is an extended and feature-enriched version of the original Recipes Dataset: 64K Dishes (from Kaggle — a high-quality collection of 64,000+ structured recipes).

Transformation Steps:

1. **Data Sanitization:** From Raw CSV to Structured Objects e.g.

"['Salt', 'Pepper!']" → Clean List: ['salt', 'pepper']

Dropping nulls and duplicates prevents any empty recipes from being recommended.

2. **Feature Engineering:** Turning Recipes into Vectors e.g.
Ingredients + Cuisine + Taste = Combined_Feature

Useful For:

- TF-IDF Vectorization
- Multi-Label Encoding
- Ordinal Mapping

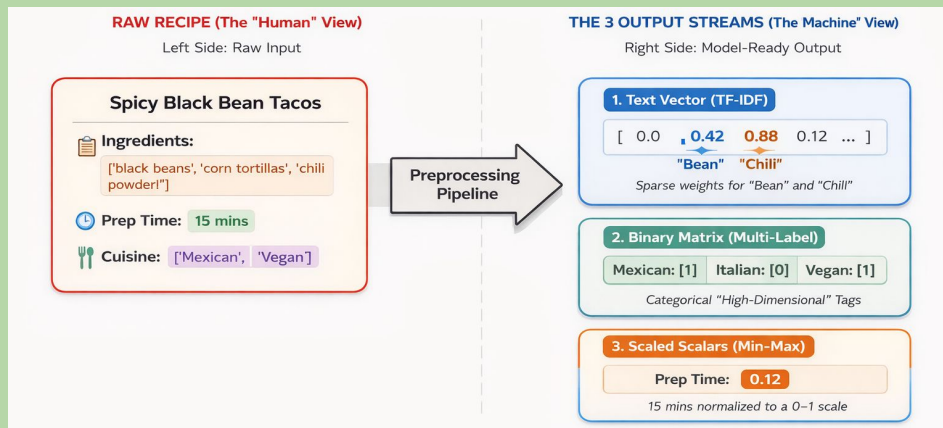
3. **Scaling & Integration:** Model Readiness & Export **Useful**

For:

- Apply MinMax Scaler
- The training and testing split
- Artifact Export (Saving .pkl files)

OUTPUT (Model-Ready Artifacts):

1. Cleaned Metadata (.csv)
2. The Feature Matrix
3. Fitted Encoders (.pkl)



Model

Data Preprocessing for Recipe Recommendation System

Need for Preprocessing

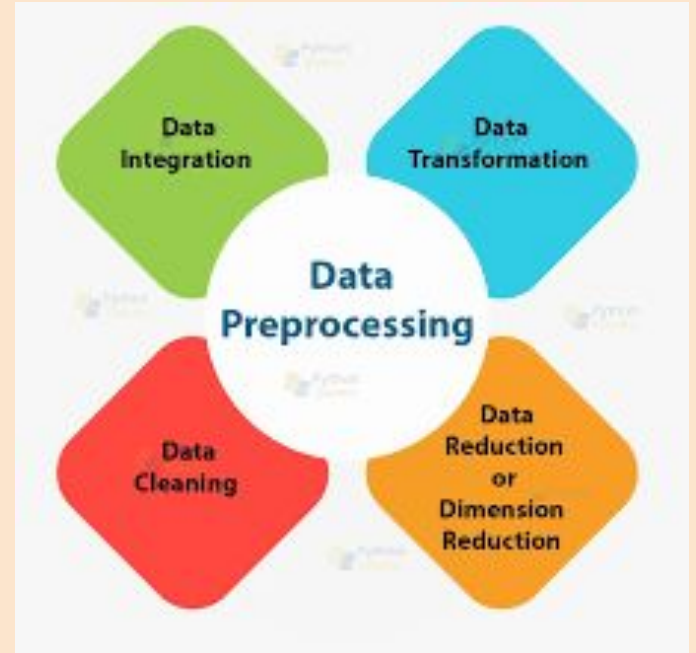
- Objective: Prepare raw recipe data for machine learning
- Input: recipes_extended.csv
- Output: recipes_cleaned.csv

Issues with Raw Data

- Ingredients stored as strings instead of lists
- Inconsistent text (uppercase, missing values)
- Presence of irrelevant columns

Goal

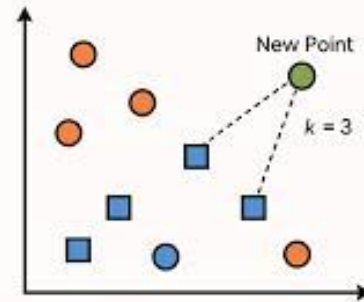
- Clean and standardize data
- Convert into a machine learning–friendly format



The Model

- A content-based recipe recommender
- Takes a list of ingredients, a diet, and taste preferences
- Returns the recipes most similar to what the user has
- No neural network
 - uses classical ML (TF-IDF + Nearest Neighbors)

K-Nearest Neighbors (KNN)



How the Model is Built

- Each recipe is converted into a numerical vector using TF-IDF
- TF-IDF weights ingredients by how unique they are to each recipe
- All recipe vectors are indexed with a Nearest Neighbors model
- The trained model is saved so it can be reused without retraining



How the Model is Used

- User ingredients are turned into a query vector with the same TF-IDF
- Recipes are filtered by dietary preference (vegan, gluten-free, etc.)
- Recipes are filtered by up to two taste preferences
- Cosine similarity finds the closest matches to the query
- The top-N most similar recipes are returned

Impact

- Addresses food waste on the individual-level
- Economic and environmental impacts
- Convenience



Tools Used

- Python
 - main programming language
- scikit-learn
 - TF-IDF vectorizer and Nearest Neighbors model
- pandas
 - loading and handling the recipe data
- Streamlit
 - web interface for trying the model
- Kaggle
 - source of the public recipe dataset
- GitHub
 - code hosting and version control

Future Work

Implementation of User Feedback as Input into Model

- Take user opinion on recipes
 - “Liked it!”, “Disliked it!”, “Hated it!”
- Feed user preference information as input into the model
 - Can be implemented on a recipe basis
 - Can alternatively be implemented on an ingredient basis
- “Anti-Preferences”
 - Take into account of undesired or unavailable ingredients
 - Take into account of undesired tastes preferred
- Live-search
 - Query/Scrape the internet for recipe data and fit it into the model

Thank You!

Sources

Real World Application

[The Environmental Impact of Food Waste | Move For Hunger](#)

[Estimating the Cost of Food Waste to American Consumers](#)

[Reducing Wasted Food | Rescuing Leftover Cuisine](#)

[U.S. Population \(1950-2025\)](#)

Scikit-learn Documentation

<https://scikit-learn.org/stable/>

TF-IDF Vectorizer

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

Nearest Neighbors

<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.NearestNeighbors.html>

Sources

Web References

1. Kaggle Datasets – <https://www.kaggle.com>
2. Scikit-learn Documentation – <https://scikit-learn.org>
3. Streamlit Documentation – <https://docs.streamlit.io>
4. Python Official Site – <https://www.python.org>
5. GitHub – <https://github.com>